

Chapter 1 Practice Assessment

Math 90-S61, Fall 2026, Jack Love

Last name, first name, ID#: _____

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

Here is a list of formulas you may find useful.

- The average rate of change of a function f on an interval $[a, b]$ is

$$AV_{[a,b]} = \frac{f(b) - f(a)}{b - a}$$

- An equation for a line with slope m containing point (x_0, y_0) is

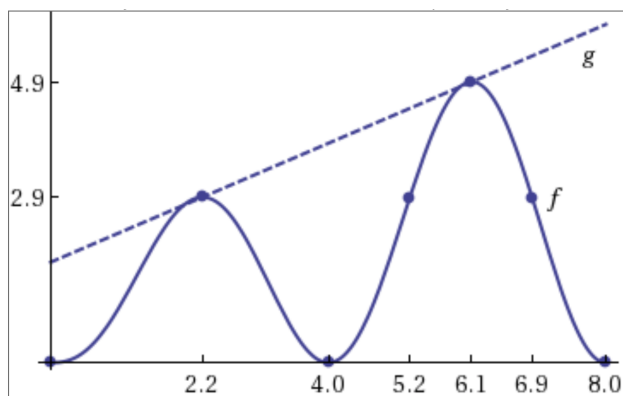
$$y = y_0 + m(x - x_0)$$

- A quadratic equation in standard form $q(x) = ax^2 + bx + c$ has

$$\text{zeros: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad y\text{-intercept: } c \quad \text{vertex: } \left(-\frac{b}{2a}, q\left(-\frac{b}{2a}\right) \right)$$

- A quadratic equation in vertex form $q(x) = a(x - h)^2 + k$ has vertex (h, k) .
- A quadratic equation in factored form $q(x) = a(x - r)(x - s)$ has zeros $x = r$ and $x = s$.

1. The image shows the graphs of $f(x)$ and $g(x)$.



- (a) Find $f(5.2) =$
- (b) Fill in the blanks in each of the two points below to correctly complete the coordinates of two points on the graph of $g(x)$.

(6.1,) (, 2.9)

- (c) For what values of x is/are $f(x) = 2.9$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x =$

- (d) For what values of x is/are $f(x) = g(x)$? If there is more than one value, write your answer as a comma-separated list of numbers.

$x =$

- (e) What is the average rate of change of $f(x)$ on the interval $2.2 \leq x \leq 6.1$?

2. A national park records data regarding the total fox population F over a 12 month period, where $t = 0$ means January 1, $t = 1$ means February 1, and so on. Below is the table of values they recorded:

t, month	0	1	2	3	4	5	6	7	8	9	10	11
F, foxes	150	143	125	100	75	57	50	57	75	100	125	143

- (a) Is t a function of F ? (Circle one)

Yes

No

- (b) Let $g(t) = F$ denote the fox population in month t . Find all solution(s) to the equation $g(t) = 125$. If there is more than one solution, give your answer as a comma-separated list of numbers.

$t =$

3. An open box is to be made from a flat piece of material 20 inches long and 6 inches wide by cutting equal squares of length x from the corners and folding up the sides.

- (a) Write the volume V of the box as a function of x . Leave it as a product of factors, do not multiply out the factors.

- (b) Write the domain of the volume function V as an open interval in the form (a, b) .

4. Find the average rate of change of $f(x) = 2x^2 - 5$ between each of the pairs of points below.

(a) Between $(3, 13)$ and $(6, 67)$. Simplify your answer as a whole number.

(b) Between (b, n) and (r, t) .

(c) Between $(x, f(x))$ and $(x + h, f(x + h))$.

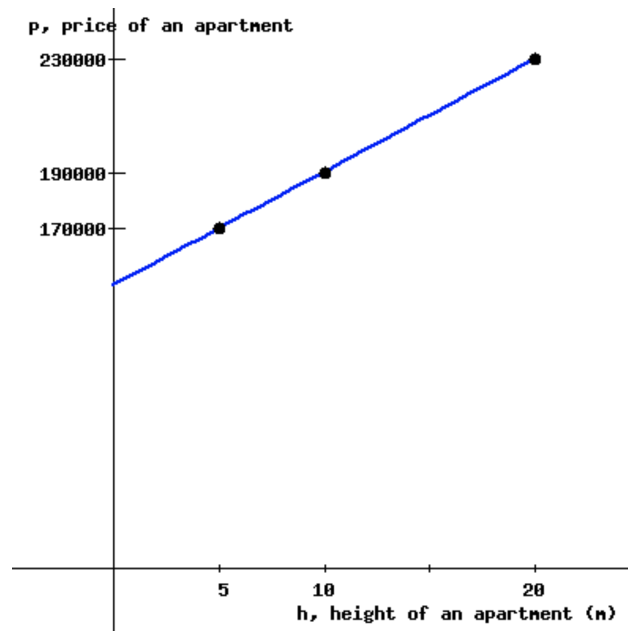
5. Let $f(x) = 36 - x^2$. Compute the average rate of change on the following intervals.

(a) $[0, 4]$

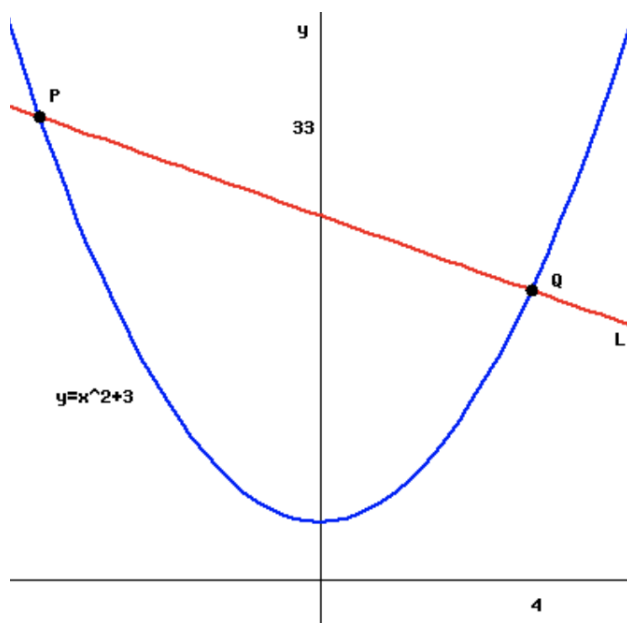
(b) $[4, 6]$

(c) $[0, 6]$

6. Find an equation in point-slope form for the linear function graphed below. Write your solution as $p = f(h) = \dots$.



7. Find the equation for the line L (graphed in red) in the figure below. The x -coordinate of the point Q is 4, the y -coordinate of the point P is 33, and the parabola (graphed in blue) has equation $y = x^2 + 3$. Leave any radicals in radical form; do not approximate.



8. Find the vertex, x -intercept(s) (if any), and y -intercept of the quadratic function

$$f(x) = x^2 - 5x - 24.$$

9. Find the zeros (if any) of the quadratic function

$$y = 4(x + 8)^2 - 8.$$

Leave any radicals in radical form; do not approximate.

10. Given that $f(x) = 5x - 6$ and $g(x) = 2x - 2$, calculate the following.

(a) $(f \circ g)(3)$

(b) $(g \circ f)(3)$

(c) $(f \circ f)(3)$

(d) $(g \circ g)(3)$

11. Let the functions f and g be given by the following table.

x	0	1	2	3	4
$f(x)$	6	4	3	4	6
$g(x)$	1	3	0	4	2

(a) Find $g^{-1}(0)$

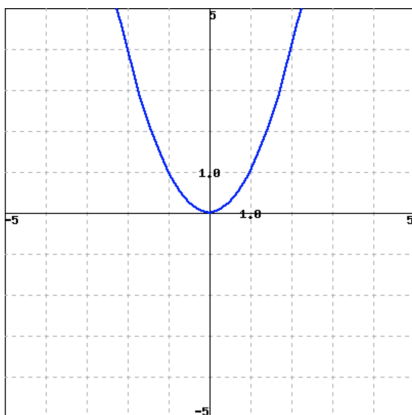
(b) Find $g(g^{-1}(1))$

(c) For what value(s) of x is $f(g(x)) = 6$?

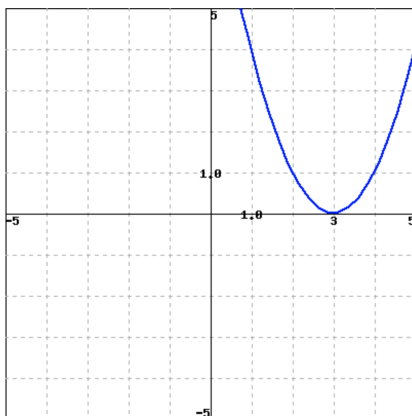
12. The cost (in dollars) of producing x air conditioners is $C = g(x) = 560 + 40x$. Find a formula for the inverse function $g^{-1}(C)$.

$$g^{-1}(C) =$$

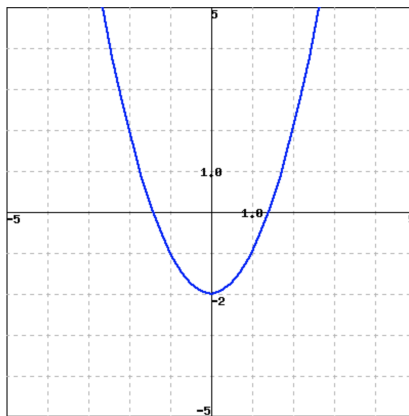
13. The graph of $y = x^2$ is given below.



Find a formula for each of the transformations whose graphs are given below.

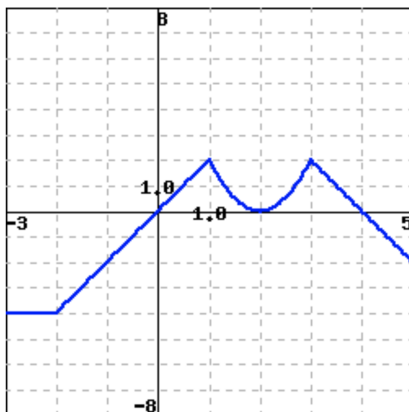


$$y =$$



$$y =$$

14. The figure below is the graph of a function $m(t)$.



Let $n(t) = m(t) + 2$, $k(t) = m(t + 1.5)$, $w(t) = m(t - 0.5) - 2.5$, and $p(t) = m(t - 1)$. Find the values of each of the following:

- (a) $n(-2.5) =$
 - (b) $k(3) =$
 - (c) $w(1.5) =$
 - (d) $p(1) =$
15. The graph of $f(x)$ contains the point $(9, 4)$. What point must be on each of the following transformed graphs?
- (a) The graph of $f(x - 6)$ must contain the point _____
 - (b) The graph of $f(x) - 5$ must contain the point _____
 - (c) The graph of $f(x + 2) + 7$ must contain the point _____

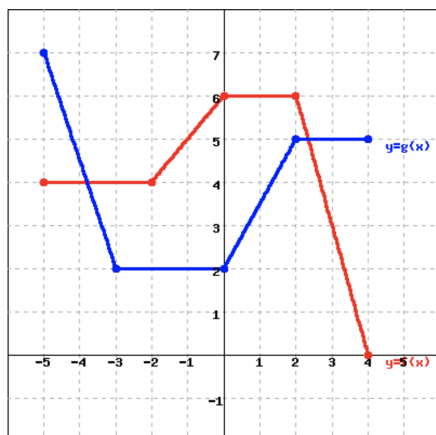
16. For $f(x) = 2 + x^2$ and $g(x) = 2x - 1$, find:

(a) $(f \cdot g)(-3) =$

(b) $(f - g)(2)$

(c) $\left(\frac{f}{g}\right)(0)$

17. In the figure below, the graph of f is shown in red and the graph of g is shown in blue. Use the graphs to evaluate each quantity given below.



(a) $f(-1) =$

(b) $g(-1) =$

(c) $(f + g)(-1) =$